Federated Learning Performance and Efficiency Analysis on Remote Sensing Datasets

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Abstract

Federated Learning (FL) allows several clients to work together to train machine learning models while keeping raw data on their devices. This feature makes FL appealing for applications that require privacy. This project creates a federated training framework that uses a compact four-layer Convolutional Neural Network (CNN) and a weighted averaging method across three popular remote-sensing datasets: EuroSAT, UC Merced, and NWPU-RESISC45. Each client trains on its own data partition, which simulates non-IID splits, and sends only model weight updates to a central server. The server combines these updates based on the sizes of the client datasets and sends back the updated model for the next rounds. We assess the framework using accuracy, macro F1-score, AUC, and total communication cost.

Experimental results show quick progress on EuroSAT, achieving a final accuracy of 0.88 in 20 rounds. Learning remained steady on NWPU-RESISC45 and UC Merced, with final accuracies of 0.74 and 0.76 to 0.79, respectively. AUC values approached 0.98 to 0.99 across the datasets. The lightweight CNN helps lower communication costs, while FedAvg offers a straightforward and effective aggregation method for different client data distributions. The study reveals the trade-offs between model complexity, communication costs, and final accuracy. It shows that federated methods can achieve strong performance in remote-sensing classification while keeping data local. The project sets a practical foundation for future research on improving aggregation, privacy measures, and model compression for edge deployment.

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1. Introduction

Remote sensing plays a key role in environmental monitoring, land-use mapping, and disaster management. As satellite and aerial imagery becomes more available, automated classification methods are crucial for extracting useful information from large amounts of data. Early methods used handcrafted features and traditional machine learning algorithms, such as SVM and random forests; these worked fairly well on small, uniform datasets. However, the complexity and variance of modern remote-sensing imagery, which includes differences in sensor types, resolutions, seasonal changes, and geographic diversity, have led to the use of deep learning, especially Convolutional Neural Networks (CNNs). These networks automatically learn hierarchical spatial features from raw pixels.

Several benchmark datasets, like UC Merced, EuroSAT, and NWPU-RESISC45, have sped up progress in scene classification using CNNs. For instance, early deep models adapted from image classification tasks, such as VGG and ResNet variants, made notable gains in accuracy by learning strong texture and shape representations. Despite these improvements, most high-performing solutions depend on having centralized access to labeled training data. Collecting remote-sensing data on a central server can be difficult or not desirable; data owners, such as regional agencies, private companies, or edge sensors, may hesitate to share raw imagery due to privacy, licensing, or bandwidth issues. Centralized collection also creates a single point of failure and raises communication and storage costs for massive image sets.

Federated Learning (FL) solves these problems by allowing collaborative training without transferring raw data. In FL, clients—including institutions, edge devices, or data silos—train locally and only share model updates, which are then combined to create a global model. Using FL in remote sensing is promising because many data collectors, like satellite operators and regional centers, already have local repositories. FL supports collective model improvement while keeping data local. This project looks into how effective a lightweight CNN is when trained in a federated setting across EuroSAT, UC Merced, and NWPU-RESISC45, focusing on convergence, fairness for each client, and trade-offs in communication.

2. Literature Review

Classical remote-sensing classification used handcrafted descriptors like color histograms, texture measures, and local features. These were combined with classifiers such as SVMs. While these methods worked well on small datasets, they needed manual feature engineering and often struggled with complex scenes. The introduction of CNNs changed this situation. Krizhevsky et al. (2012) showed how powerful deep convolutional architectures could be for image recognition. This inspired many adaptations for remote sensing. Later work applied pretrained ImageNet models and fine-tuning strategies to aerial imagery, leading to significant improvements in accuracy and reliability.

Notable contributions in remote sensing include the UC Merced dataset benchmark studies (Yang and Newsam, 2010) and EuroSAT (Helber et al., 2019), which highlighted the importance of multispectral data for land-use classification. The NWPU-RESISC45 dataset emphasized diversity in large-scale scenes. Researchers have used architectures from the VGG and ResNet families, along with transfer learning and data augmentation, to improve performance on these datasets. However, these studies typically assume

centralized training and full access to labeled data.

Federated Learning was introduced by McMahan et al. (2017) using the FedAvg algorithm. This method allows for efficient decentralized model training. Later works (Bonawitz et al., MLSys 2019) focused on scalability and secure aggregation for real-world applications. Recent FL research looks into personalization, compression, and strong aggregation to tackle these issues.

Despite these advancements, there is little literature on FL for remote sensing. Existing studies either apply FL to single, small datasets or focus on system-level improvements without thoroughly analyzing how well classification performs for each client on large remote-sensing benchmarks. Key issues in previous work include: (1) dependence on centralized labeled data or limited evaluations across sites, (2) not considering the trade-offs between communication costs and model accuracy for high-resolution images, and (3) not exploring model compression and privacy mechanisms in remote sensing enough. This project aims to fill these gaps by evaluating a compact CNN with FedAvg across three different remote-sensing datasets. It will analyze convergence behavior, fairness for each client, and communication costs.

3. Dataset Description

3.1. Datasets Overview

Table 1: Dataset summary

Dataset	**Classes**	**Images**	**Resolution**	**Notes**
NWPU-RESISC45	45	31,500	256×256	Highly diverse scenes
EuroSAT	10	27,000	64×64	Sentinel-2 RGB used
UC Merced	21	2,100	256×256	Small but high-quality

3.2. Key Points

- NWPU contains high intra-class variation and many scene types.
- EuroSAT offers multispectral signals; experiments use RGB bands for consistency.
- UC Merced is compact and suitable for rapid experimentation.

4. System Architecture

The federated learning system in this project uses a standard server-client architecture. It allows for collaborative model training without centralizing data. A central server initializes the global CNN model and sends it to all participating clients. After local training, it receives their updated weights and combines these updates using the FedAvg algorithm. Each client has a private, non-IID subset of the dataset and trains locally on its own. Only the model parameters go back to the server, which ensures that the

raw remote-sensing images stay on the client side. This setup enables learning from geographically diverse and sensitive datasets while reducing bandwidth use and protecting privacy. The communication between the server and clients happens over several rounds, gradually enhancing the global model until it converges.

5. Working Process

1. **Data Preparation:** Images from all datasets are resized to 224×224, normalized, and lightly augmented (flips and rotations) to improve generalization and maintain uniform input format.

2. **Client Data Distribution:** The datasets are split among clients in a non-IID manner, where each client receives a different class distribution or unequal quantities of data, closely representing real-world scenarios.

3. **Model Initialization:** A lightweight four-layer CNN is initialized on the server and the global model weights are broadcast to all participating clients at the start of each round.

4. **Local Client Training:** Each client trains the model locally on its private data using the Adam optimizer and cross-entropy loss, producing updated model weights without sharing raw data.

5. **FedAvg Aggregation:** Clients return their updated weights and local sample counts (n_i), and the server aggregates them using the weighted FedAvg rule to create an updated global model.

6. **Evaluation and Iteration:** After each round, the global model is evaluated using accuracy, F1-score, AUC, and communication cost. The updated model is redistributed to clients and training continues until convergence.

6. Convolutional Neural Network (CNN) Architecture

6.1. CNN Used in This Project

The lightweight CNN extracts spatial features from remote-sensing images through four convolution–pooling blocks. Early layers learn edges and textures; deeper layers capture scene-level patterns. Features are flattened and passed to a fully connected layer and an output softmax for class probabilities. Each client trains this CNN locally and transmits only model weights during aggregation.

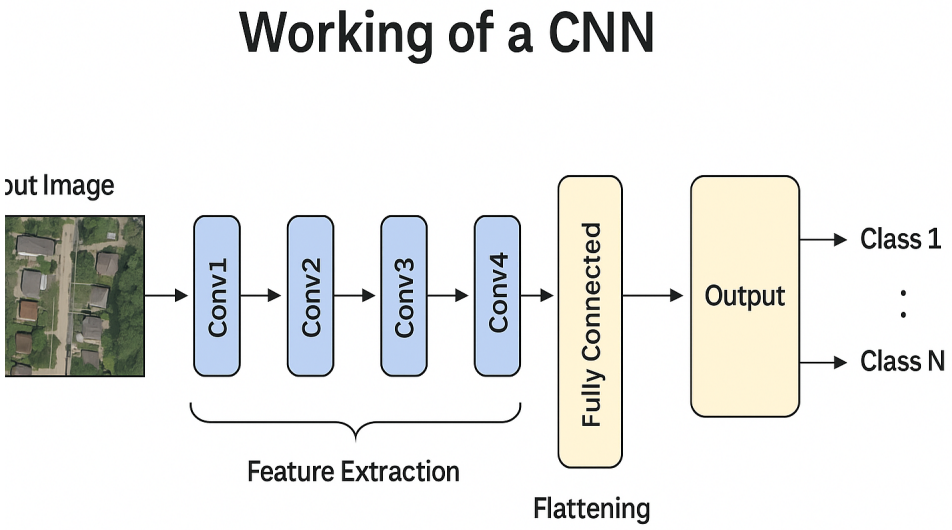

Figure 1: Working of the CNN model used in this project. Four convolutional blocks extract features followed by flattening and fully connected classification layers.

6.2. Layer Specification

Table 2: CNN layer description

Layer	Config	Output
Input	3×224×224	–
Conv1	16 filters + MaxPool	16×112×112
Conv2	32 filters + MaxPool	32×56×56
Conv3	64 filters + MaxPool	64×28×28
Conv4	128 filters + MaxPool	128×14×14
Flatten	–	25,088 features
FC1	512 units	512
Output	C units	C

6.3. Parameter Summary

Table 3: Trainable parameters across datasets

Dataset	Classes	Parameters
NWPU-RESISC45	45	12,966,093
EuroSAT	10	12,948,138
UC Merced	21	12,953,781

7. Federated Learning Aggregation (FedAvg)

FedAvg computes the weighted average of client updates:

$$W_{global} = \sum_{i=1}^{K} \frac{n_i}{\sum_{j=1}^{K} n_j} W_i,$$

where n_i is the number of samples on client i. This gives larger contribution to clients with more data while maintaining decentralization.

8. Experimental Setup

- Python 3.8, PyTorch, NumPy, scikit-learn
- Learning rate: 0.001, Optimizer: Adam
- Batch size: 16–32; Local epochs: 1–5
- Communication rounds (example): EuroSAT 20, UC Merced 28, NWPU 92

9. Results and Analysis

9.1. EuroSAT (20 Rounds)

Accuracy improved from 0.21 to 0.88; F1-score from 0.12 to 0.88; AUC from 0.78 to 0.99. Client accuracies ranged from 0.754 to 0.885. Communication cost reached 20 GB.

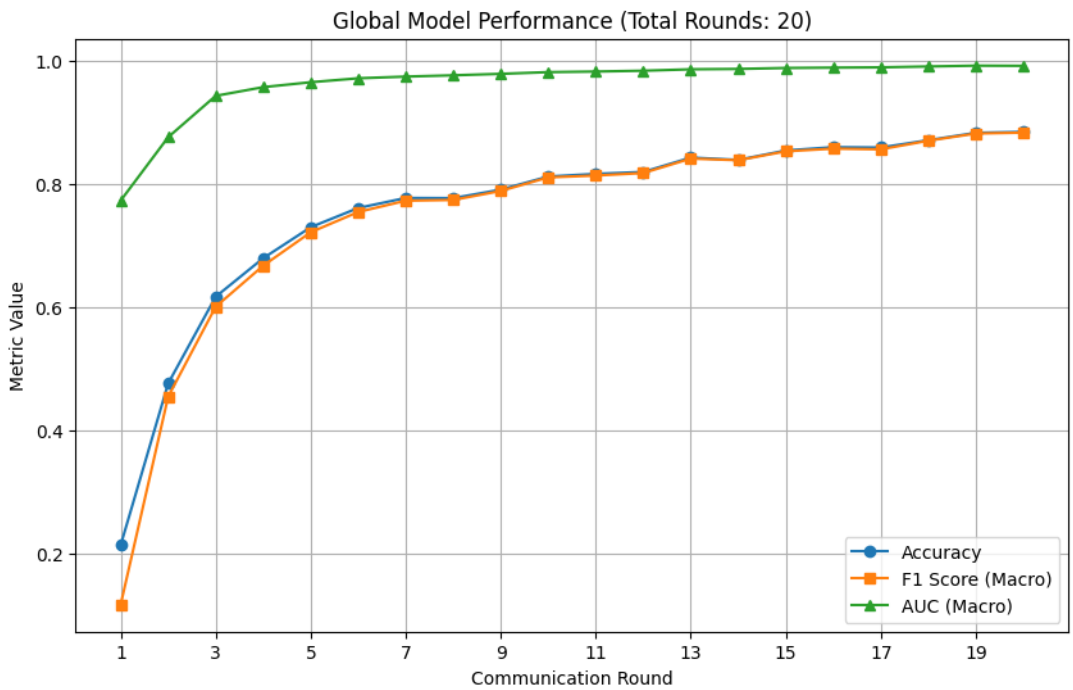

Figure 2: EuroSAT: Accuracy, F1 and AUC over rounds.

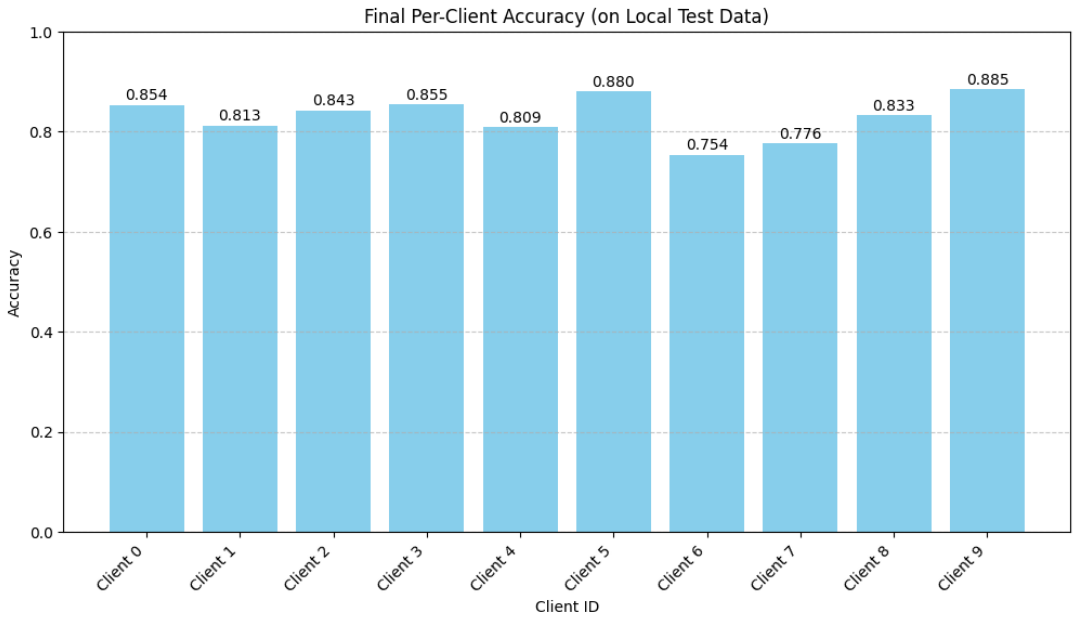

Figure 3: EuroSAT: Per-client accuracy.

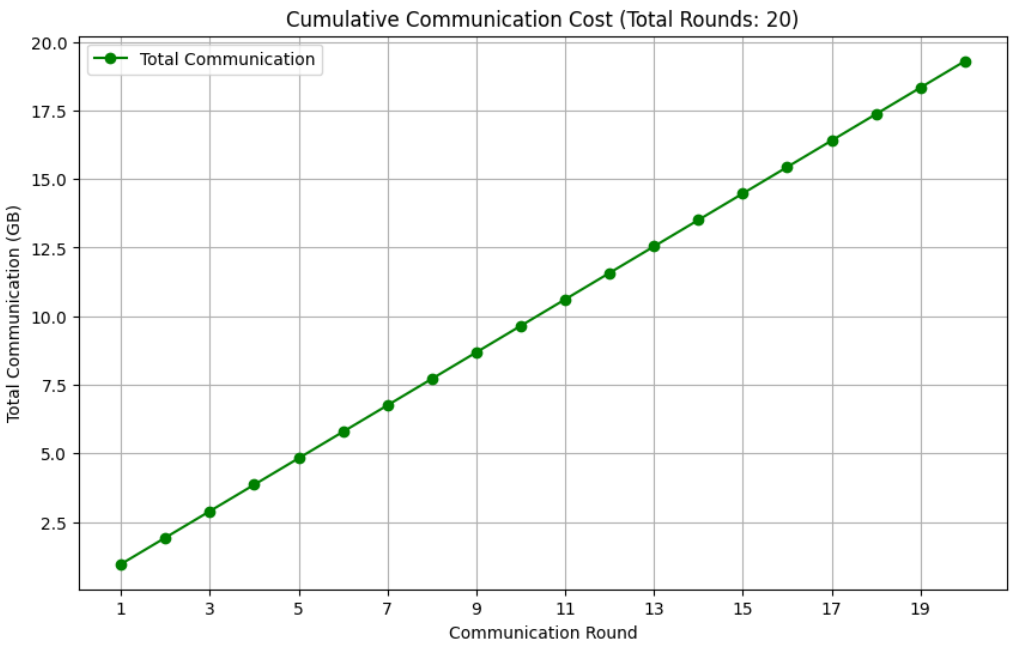

Figure 4: EuroSAT: Communication cost.

9.2. NWPU-RESISC45 (92 Rounds)

Accuracy reached 0.73–0.74; F1 0.74; AUC 0.98–0.99. Client accuracies ranged 0.763–0.827. Communication cost 90 GB.

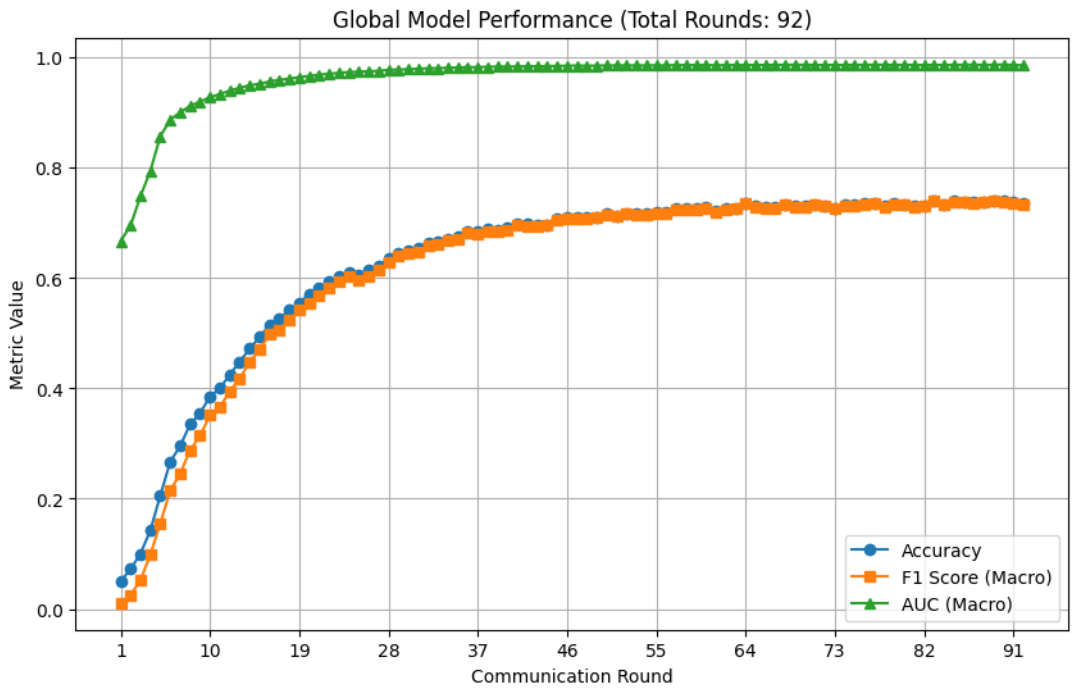

Figure 5: NWPU: Accuracy, F1 and AUC over rounds.

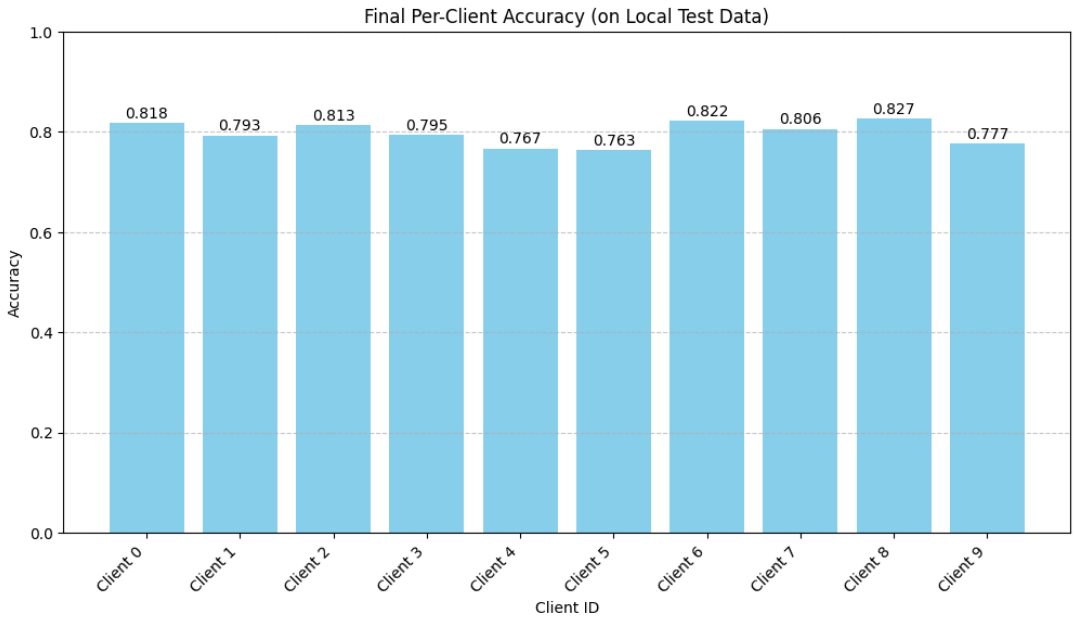

Figure 6: NWPU: Per-client accuracy.

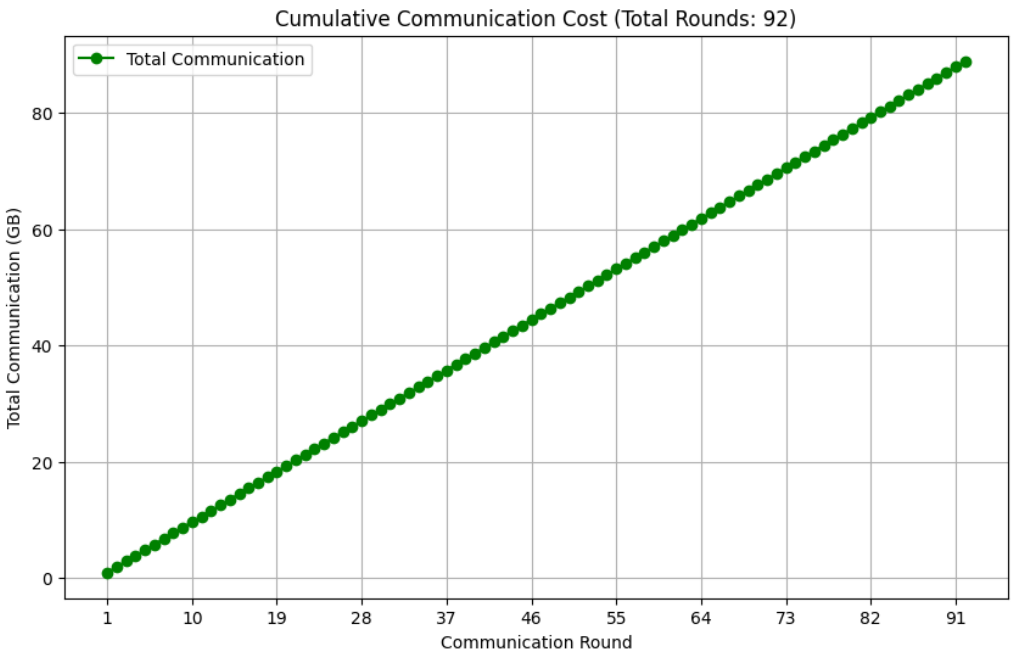

Figure 7: NWPU: Communication cost.

9.3. UC Merced (28 Rounds)

Global accuracy: 0.76–0.79; F1 0.74; AUC 0.98–0.99. Client accuracies: 0.6716–0.8209. Communication cost: 13 GB.

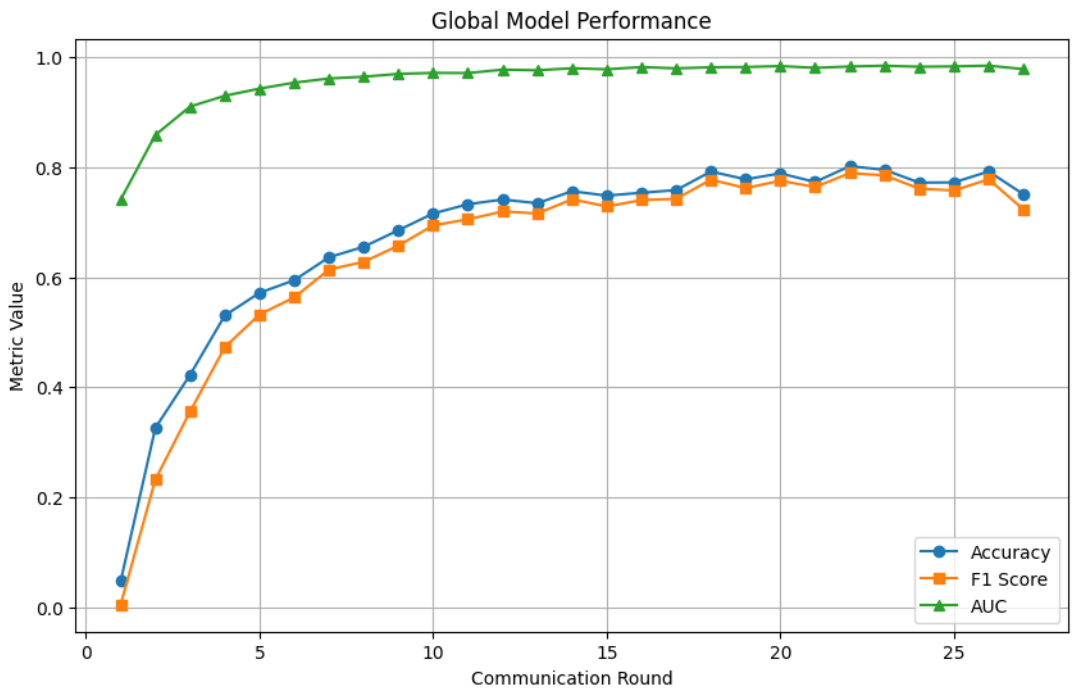

Figure 8: UC Merced: Global model metrics across rounds.

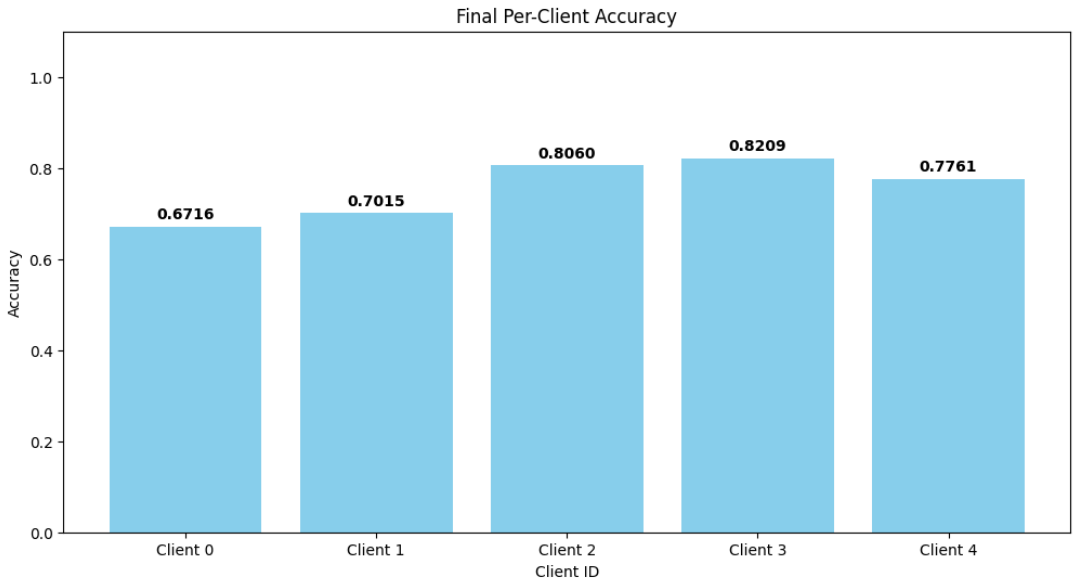

Figure 9: UC Merced: Per-client accuracy.

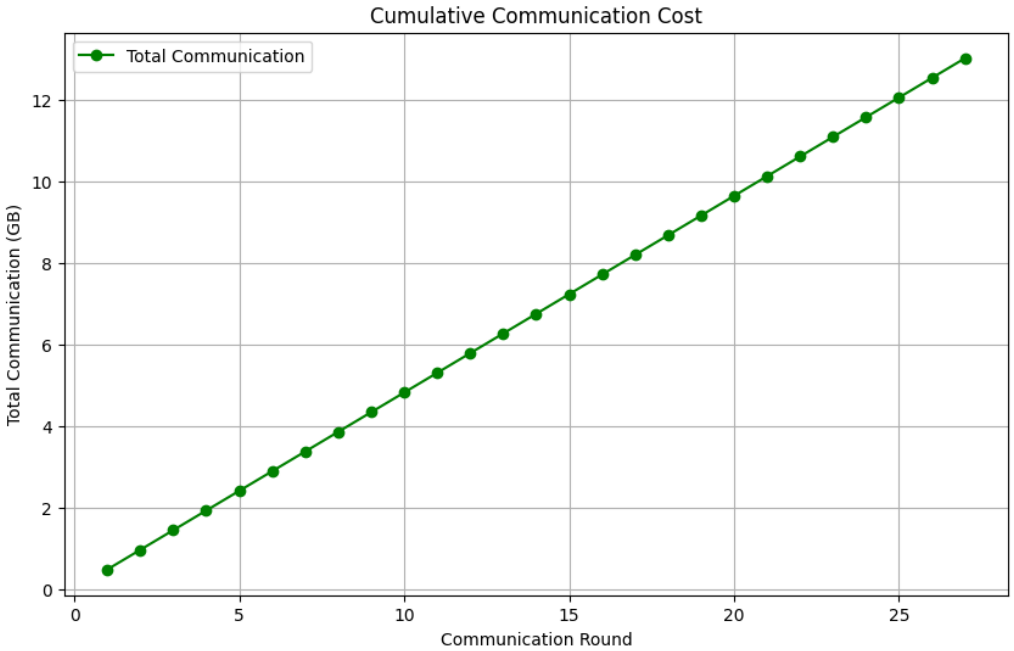

Figure 10: UC Merced: Communication cost.

10. Conclusion

This project demonstrates that federated learning with a lightweight CNN can effectively learn from distributed remote-sensing data while preserving data locality. Across EuroSAT, UC Merced, and NWPU-RESISC45, the federated approach achieved competitive accuracies and high AUC values, with acceptable communication overhead. The results validate FL as a promising direction for collaborative remote-sensing model development when raw data sharing is constrained.

11. Future Work

Future work will focus on improving model performance under non-IID conditions by integrating FL algorithms like FedProx, FedNova, FedOpt, and Scaffold. We also plan to expand the study by training on more remote-sensing datasets to test scalability in different environments. Additionally, the current lightweight CNN can be replaced with a deeper or more complex architecture to capture richer spatial features and attain higher classification accuracy.

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